Supplementary Materials

What scale buys for single-cell foundation models: a controlled probe of Geneformer V1, V2-104M, and V2-316M

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1. Per-comparison paired statistics (Supplementary Table S1)

This table reports every paired contrast referenced in the main text. Pairing units differ by probe: L1 and L3 are paired across the five donor folds ($n = 5$); L2 is paired across regulons $\times$ folds ($n = 375$, one paired observation per TF $\times$ fold combination retained after vocabulary filtering and minimum-positive-count thresholding). Δ is the mean paired difference (row 1 $-$ row 2). The 95% CI is a 2,000-iteration percentile bootstrap over paired differences (seed 0). p is the two-sided Wilcoxon signed-rank statistic. With $n = 5$ the smallest attainable two-sided Wilcoxon p is $2 \times (1/2)^5 = 0.0625$; a value of 0.0625 indicates the strongest possible non-tied paired-rank evidence the test can produce at this sample size, not a marginal result.

Table 1: Per-comparison paired statistics for every contrast referenced in the main text. Sign convention: Δ = (row A) $-$ (row B); positive values favor row A. Wilcoxon p-values are two-sided. "random-init" denotes the matched-dimension geometry-falsifying control (256-d for V1-10M, 768-d for V2-104M, 1152-d for V2-316M). "neg=random" denotes the L2 sensitivity variant where TF–non-target negatives are sampled from the full vocabulary rather than matched within the same regulon's empirical degree distribution.

Probe	**Comparison (A − B)**	n	Δ	**CI low**	**CI high**	p
L1 (cell-type)	V2-104M − V1-10M	5	0.0765	0.0402	0.1172	0.0625
L1 (cell-type)	V2-316M − V1-10M	5	0.0700	0.0472	0.0927	0.0625
L1 (cell-type)	V2-316M − V2-104M	5	−0.0065	−0.0342	0.0171	1.000
L1 (cell-type)	V2-316M − bag-of-genes	5	0.0192	−0.0011	0.0425	0.312
L1 (cell-type)	V2-104M − bag-of-genes	5	0.0257	−0.0100	0.0632	0.438
L1 (cell-type)	V1-10M − bag-of-genes	5	−0.0508	−0.0575	−0.0439	0.0625
L2 (TF→target)	V2-104M − V1-10M	375	0.0545	0.0407	0.0679	2.4×10^{-16}
L2 (TF→target)	V2-316M − V1-10M	375	0.0513	0.0380	0.0657	3.2×10^{-15}
L2 (TF→target)	V2-316M − V2-104M	375	−0.0032	−0.0138	0.0083	0.447
L2 (TF→target)	V2-316M − bag-of-genes	375	0.1670	0.1516	0.1830	4.4×10^{-50}
L2 (TF→target)	V2-104M − bag-of-genes	375	0.1702	0.1538	0.1869	6.8×10^{-48}
L2 (TF→target)	V1-10M − bag-of-genes	375	0.1157	0.0992	0.1319	1.7×10^{-34}
L2 (TF→target, neg=random)	V2-316M − V1-10M	375	0.0149	0.0033	0.0268	0.0111
L3 (hub identity)	V2-104M − V1-10M	5	−0.0090	−0.0551	0.0264	1.000
L3 (hub identity)	V2-316M − V1-10M	5	−0.0199	−0.0532	0.0131	0.625
L3 (hub identity)	V2-316M − V2-104M	5	−0.0109	−0.0284	0.0066	0.438

Table S1 (continued)

Probe	Comparison (A − B)	n	Δ	CI low	CI high	p
L3 (hub identity)	V1-10M − bag-of-genes	5	0.2192	0.1319	0.2920	0.0625
L3 (hub identity)	V2-104M − bag-of-genes	5	0.2102	0.1376	0.2728	0.0625
L3 (hub identity)	V2-316M − bag-of-genes	5	0.1993	0.1332	0.2652	0.0625
L3 (hub identity)	V1-10M − V1-10M random-init	5	0.2935	0.2423	0.3637	0.0625
L3 (hub identity)	V2-104M − V2-104M random-init	5	0.1790	0.1426	0.2154	0.0625
L3 (hub identity)	V2-316M − V2-316M random-init	5	0.2556	0.2136	0.2966	0.0625

2. Notes on interpretation

Why $p = 0.0625$ recurs at $n = 5$. The two-sided Wilcoxon signed-rank test on five paired observations cannot produce a p-value below $2 \times (1/2)^5 = 0.0625$ when all paired differences share a sign. Every $p = 0.0625$ row in this table is therefore the smallest-attainable Wilcoxon p on five donor folds, not a near-miss. The companion 95% bootstrap CI is the more informative effect-size summary at this sample size, and is the reason every per-comparison interpretation in the main text is anchored on the CI rather than p alone.

Why the L3 hub-identity gap is reported as the strongest finding. Three trained-vs.-random-init contrasts at L3 (V1-10M, V2-104M, V2-316M) each achieve the floor $p = 0.0625$ with non-overlapping CIs around $\Delta \approx 0.18$–0.29. Treating the three random initializations as independent (they are, by construction — different seed, different draw of weights at each scale), the joint sign-test probability of all three trained-versus-random contrasts pointing the same direction under H_0 is $(1/2)^3 = 0.125$, and the joint probability that all three exceed $\Delta = 0$ by more than the smallest observed CI lower bound (here 0.1426) is substantially smaller. The main-text interpretation deliberately walks back to $(1/2)^3$ rather than aggregating across the trained-versus-bag-of-genes rows because those latter rows share the same trained AUCs across model scales (within-row dependence) and the same bag-of-genes baseline across the three scale rows (cross-row dependence), so they cannot be combined as independent sign tests.

L1 prose convention. The main text reports L1 as “fails to exceed bag-of-genes” rather than “matches” or “ties” bag-of-genes. The data here support that framing: the V2-316M − bag-of-genes contrast at L1 has $\Delta = +0.019$ with CI $[-0.001, +0.043]$ and Wilcoxon $p = 0.31$, which fails to reject the null but does not constitute a formal equivalence test against any pre-specified equivalence margin.